

Verifier-Assisted versus One-Shot LLM Intent→Configuration: A Paired NetOpsTraceBench Study

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Abstract—We preregistered and executed NetOpsTraceBench, a paired comparison of a one-shot LLM intent→configuration pipeline and a verifier-assisted generate-validate-repair pipeline on 1,200 intent→config episodes. Using gpt-5-mini, a deterministic NetOps validator, and Mininet-style emulation, we applied a preregistered binary episode success rule (validator accepts final config AND emulator shows no availability degradation above tolerance AND no automatic rollback). Of 1,200 planned paired tasks we completed 1,199 pairs. Baseline success = 1,196/1,199 (0.9974979149); guarded success = 1,199/1,199 (1.0000000000). The paired-success-rate difference (guarded - baseline) = 0.0025020851 (95% bootstrap percentile CI [0.0000, 0.0058381985], bootstrap_seed=1729, resamples=10,000); exact McNemar two-sided $p = 0.25$. The preregistered power target sought a 5 percentage-point minimum detectable effect; given the observed near-ceiling baseline, the study lacked power to detect much smaller absolute gains. Representative validator traces and provider audit logs show repair was invoked only three times. Under these controlled emulation and task-corpus conditions we find no statistically significant advantage for the verifier-assisted pipeline; we report artifacts, analytic deviation (Wilcoxon → exact McNemar for binary outcomes), and detailed execution accounting to support reproducible interpretation.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate operator intent expressed in natural language into executable device configurations for network operations (NetOps). Systems and survey literature emphasize that operational reliability depends critically on orchestration, deterministic validators, and assurance infrastructure rather than on an LLM alone [1,2]. Generate-validate-repair (verifier-assisted) architectures are a canonical mitigation strategy: an LLM proposes candidate outputs, an automated verifier checks constraints and safety, and an automated repair loop iterates until the candidate satisfies verifier rules [3–5]. Despite conceptual support, publicly archived, preregistered, paired evidence measuring whether verifier assistance yields an operationally meaningful improvement (rollback-aware binary success) relative to one-shot generation is sparse.

We address this gap with NetOpsTraceBench: a preregistered, paired, emulator-grounded study comparing (A) baseline one-shot NL→config generation and (B) a guarded generate-validate-repair pipeline using the same shared initial candidate. The study asked: Can a reproducible, workflow-centered benchmark of paired intent→config episodes produce an objective paired binary success metric that reliably distinguishes verifier-assisted pipelines from one-shot generation across two

simulated topologies? Our contributions are (1) a preregistered experimental design and full execution artifacts (study ID rX4f7-1); (2) a paired analysis on 1,199 completed pairs using gpt-5-mini and a deterministic validator; (3) transparent reporting of an analytic deviation (final confirmatory test used an exact McNemar test appropriate for paired binary outcomes) and execution accounting that explains why the guarded pipeline rarely required repair.

II. RELATED WORK

Recent surveys and system papers position LLMs within AIOps/NetOps pipelines and emphasize the importance of verifiers, provenance, and end-to-end workflow evaluation [1,6]. Work on verifier-guided NL→formal translations and intent→topology compilation has shown the feasibility of generate-validate-repair approaches in domains such as security topology compilation and CTL/specification generation [3,7]. Empirical and conceptual studies document LLM failure modes relevant to NetOps—hallucination, incorrect semantics/units, overconfidence, provenance loss, and prompt injection—which motivate guarded architectures and runtime auditing [8–10]. Several recent prototypes demonstrate LLM-assisted configuration generation and misconfiguration detection but typically evaluate on constructed or small held-out sets [4,5,11]. That literature motivates workflow-centered evaluation metrics (trace quality, rollback measurement, canaries) rather than one-shot QA metrics; NetOpsTraceBench operationalizes a paired, rollback-aware binary success metric in a preregistered framework and archives the run artifacts for reproducibility.

III. METHODOLOGY

Preregistration and scope: The study preregistration (rX4f7-1; preregistration SHA-256 = 77abe8b493bde44484e0b3d02cfd7f9ec1621ff7e7847889082ae177a7ed5712) froze task selection, inclusion/exclusion rules, contamination thresholds, stopping rules, seeds, analysis plan, and failure treatments before execution. NetOpsTraceBench_v1 comprises 1,200 paired intent→config tasks split into two topology clusters (edge = 600, core = 600). Each task is public synthetic or consented with deterministic provenance recorded. Tasks are vendor-agnostic canonical sequences rendered into device configurations for emulator execution. Inclusion/exclusion rules were deterministic: pairs were excluded if contamination

checks flagged training-data overlap above thresholds or if both episodes failed to produce parsable vendor renderings after one automated re-execution attempt; exclusions are logged and reported.

Pipelines and instrumentation: We compared (A) Baseline one-shot — a single LLM call producing an initial candidate configuration; (B) Guarded verifier-assisted — the same initial candidate plus at most one automated repair iteration triggered when the deterministic validator rejected the candidate. Execution used `requested_model = gpt-5-mini` (provider recorded as `openai_responses`) orchestrated by adapter `hosted_netops_gvr_v1`. Deterministic `netops_validator_v5` handled parsing, intent-constraint checks, and emulator preflight; the task generator was `deterministic_netops_task_generator_v5`. The execution manifest records `model_calls_used = 1,203` and `maximum_model_calls = 2,400`.

Success metric and confirmatory analysis: The preregistered binary episode success rule required (1) validator acceptance of the final configuration, (2) emulator execution with no availability degradation above the preregistered tolerance, and (3) no automatic rollback during simulated execution. The preregistration specified a nonparametric paired procedure (Wilcoxon signed-rank) for paired outcomes; because realized outcomes are binary, the archived confirmatory analysis used an exact McNemar two-sided test on the paired 2×2 contingency table and computed a paired bootstrap percentile 95% CI for the paired success-rate difference using frozen `bootstrap_seed = 1729` and 10,000 resamples. This post-preregistration analytic choice is documented in archived artifacts (`analysis/paired_contingency_table.csv`, `analysis/results.json`, `analysis/analysis_log.jsonl`). The primary estimand is `paired_success_rate_difference_guarded_minus_baseline`. The preregistered power plan targeted a minimum detectable effect (MDE) of 5 percentage points (absolute) at two-sided $\alpha=0.05$ and power 0.8 given assumed baseline ≈ 0.70 ; the power rationale and sensitivity scenarios are recorded in the preregistration.

IV. RESULTS

Execution and pair accounting: The run completed 2,398 of 2,400 planned episodes and produced 1,199 complete paired observations (`completed_pairs = 1,199`; `failed_episode_count = 2`). The analysis used `complete_pair_count = 1,199`. Provider audit and orchestration logs record `hosted_model_call_event_count = 1,203` (`completed_hosted_model_call_event_count = 1,202`; `failed_hosted_model_call_event_count = 1`) and `stage_counts` indicating `shared_initial_generation = 1,200` and `repair = 3`. These counts show that a single shared initial generation was produced per task and that the guarded pipeline invoked repair calls only three times across the entire corpus.

Primary confirmatory outcomes (paired contingency and rates): The paired 2×2 contingency table (`analysis/paired_contingency_table.csv`) is: `n11 =`

`1,196` (both methods succeeded), `n10 = 3` (guarded succeeded while baseline failed), `n01 = 0` (baseline succeeded while guarded failed), `n00 = 0` (both failed). From these counts: `baseline_success_count = 1,196` (`baseline_success_rate = 1,196/1,199 = 0.9974979149291076`) and `guarded_success_count = 1,199` (`guarded_success_rate = 1,199/1,199 = 1.0`). The observed paired-success-rate difference (`guarded - baseline`) = `0.002502085070892446`.

Exact paired inference and bootstrap CI: Because confirmatory outcomes are binary and the observed data are concentrated in concordant cells, we used an exact McNemar two-sided test on the discordant pair counts (`n10 = 3`, `n01 = 0`). The archived exact McNemar two-sided $p = 0.25$. The paired bootstrap percentile 95% CI for the paired-success-rate difference (computed with `bootstrap_seed = 1729` and 10,000 resamples as recorded in `analysis/analysis_log.jsonl`) is `[0.0000, 0.005838198498748957]`. The CI lower bound reaching 0.0 reflects the high proportion of concordant successful pairs and the small number of discordant pairs; those data characteristics limit resolution for small absolute effects.

Missingness and failed calls: The preregistered missingness summary (`analysis/missingness_summary.csv`) reports `complete_pairs = 1,199`; `failed_baseline_episodes = 1`; `failed_guarded_episodes = 1`; `both_missing_pairs = 1`. The deterministic exclusion rules were not triggered for contamination; `analysis/contamination_summary.csv` is empty. The single paired exclusion implied by `both_missing_pairs = 1` is the accounting source for the one planned pair drop from 1,200 to 1,199 complete pairs.

Repair invocation diagnostics: `Stage_counts` (execution manifest and `deterministic_reconciliation.json`) show `repair = 3` while `shared_initial_generation = 1,200`. The repair count matches `repair_provider` traces present for exactly three episodes in the `execution/provider_calls` and `execution/scoring` artifacts. Scorer and validator traces (examples in `execution/scoring/*_json`) report `validation_after.valid = true` for the majority of episodes and `repair_applied = false`, indicating that initial candidates typically satisfied `deterministic_netops_validator_v5` checks and therefore required no automated repair.

Representative artifact evidence: Representative task manifest entries and scoring artifacts illustrate the evidence pattern. For example, `execution/scoring/task-000001-baseline.json` and `execution/scoring/task-000001-guarded.json` show identical `shared_initial_candidate` content and validator traces with `validation_after.valid = true` and `repair_applied = false`; difficulty for that task is labeled "medium" in the `task_manifest` entry. Examples with higher difficulty labels ("hard", "very_hard") appear in representative task entries (`task-000002` level "hard", `task-000003` level "hard") and still produced accepted shared initial candidates in many cases. These artifact-level diagnostics demonstrate (a) heterogeneous declared task difficulty across the corpus, and (b) that under the chosen deterministic task generator and validator settings the initial LLM output frequently met intent constraints.

Interpretation of discordant pattern: The observed discor-

dant pattern ($n_{10} = 3$, $n_{01} = 0$) means all discordant pairs favored the guarded pipeline; however, the absolute discordant count ($3/1,199 \approx 0.25\%$) is too small to yield statistical significance under the prespecified inferential thresholds. Under the exact McNemar framework the two-sided $p = 0.25$ (archived), reflecting the binomial probability of observing such an imbalanced discordant split given the small number of discordant pairs. The paired bootstrap CI’s narrow upper bound (≈ 0.00584) but lower bound at zero quantifies that any true absolute benefit, if present, is small in magnitude within this corpus and validator configuration.

Reproducibility anchors: All numbers above match archived artifacts: `analysis/paired_contingency_table.csv` ($n_{11}/n_{10}/n_{01}/n_{00}$), `analysis/condition_summary.csv` (per-condition counts and success rates), `analysis/results.json` (estimate, p -value, CI, bootstrap parameters), `execution/execution_manifest.json` (provider audit counts and timestamps), and representative scoring artifacts (`execution/scoring/*.json`) that record validator trace fields `validation_before/validation_after`, `repair_applied` flags, and `normalized_configuration` values. The run’s bootstrap parameters (`seed = 1729`, `resamples = 10,000`) and analysis log are included in `analysis/analysis_log.jsonl` to enable exact reproduction of the CI reported above.

V. DISCUSSION

The statistical evidence and operational diagnostics together explain why the preregistered paired comparison did not find a statistically significant advantage for the guarded pipeline in this run. The exact McNemar two-sided test ($p = 0.25$) and the paired bootstrap percentile 95% CI [0.0000, 0.0058381985] both include zero, indicating that the observed paired success-rate difference (≈ 0.25 percentage points) is consistent with sampling variability under the recorded discordant counts ($n_{10} = 3$, $n_{01} = 0$). Those discordant counts are the proximal determinant of inference here: with 1,199 complete pairs, only three pairs were discordant and all favored the guarded pipeline, which yields a point estimate but insufficient evidence to reject the null at $\alpha = 0.05$. This arithmetic — small discordant counts in an otherwise near-ceiling outcome distribution — is why the preregistered MDE of 5 percentage points (chosen under an assumed baseline ≈ 0.70) was not matched by the observed data; the study therefore lacked confirmatory power to detect such a small absolute improvement.

Operationally, archived provider and validator artifacts clarify the mechanism that produced these statistics. The provider audit reports `model_calls_used = 1,203` with `stage_counts` showing `shared_initial_generation = 1,200` and `repair = 3`, and representative validator traces (e.g., `execution/scoring/task-000001-baseline.json`) show `validation_after.valid = true` for typical episodes. Together these elements indicate that the shared initial candidates satisfied the deterministic `netops_validator_v5` checks in the overwhelming majority of episodes and so the guarded pipeline’s repair stage was rarely invoked. Thus, under the exact combination of (a) a synthetic, well-specified task generator that produces tasks

with explicit required sequences, and (b) a validator configuration that enforces those sequences in a way that treats correct initial candidates as acceptable, the marginal operational value of generate–validate–repair for the binary success metric is necessarily small.

These findings do not falsify the general utility of verifier-assisted architectures. Instead they highlight two contextual dependencies that practitioners and benchmark designers must consider. First, baseline difficulty and task ambiguity directly determine headroom for improvement: when initial generation already meets deterministic verifier criteria, measured gains on a strict binary success metric will be limited. Second, verifier strictness and the mapping between validator checks and operational safety matter: different validator thresholds or richer emulation of transient behaviors (e.g., routing convergence, vendor-specific command ordering) would change repair invocation rates and the distribution of discordant pairs. Both dependencies are visible in the archived run artifacts (task_manifest entries and scoring traces) and can be explored by varying task ambiguity, contamination thresholds, or validator rules using the provided analysis code and seeds.

Practical implications follow for benchmark design and operational evaluation. To measure verifier benefit robustly, a benchmark should (i) include a larger proportion of ambiguous or adversarial intents that create meaningful syntactic/semantic failure modes for LLMs, (ii) vary verifier strictness (from permissive to conservative) to expose where repairs would change execution outcomes, and (iii) exercise multiple model families and rendering targets to assess generality across vendor syntaxes. The preregistered sensitivity scenarios and the archived artifacts make such follow-up experiments reproducible: the run archives deterministic seeds, task generator configuration, validator traces, and analysis scripts so other researchers can replay alternative verifier sensitivities or inject adversarial perturbations without ambiguity.

Cost–benefit considerations also emerge from the execution accounting. Repair was invoked only three times, and the total model call audit ($\approx 1,203$ calls) shows a small additional invocation footprint relative to the number of pairs. In settings where repairs are rare because validators accept correct initial candidates, the operational overhead of a guarded pipeline may be modest; however, this mechanical accounting does not capture human-in-the-loop costs, potential latency requirements, or the upstream effort to build and maintain robust validators and emulators. Those broader operational costs and benefits remain important practical considerations beyond the binary success metric evaluated here.

Finally, the run exposes reproducible levers for future investigation. The archived CI, contingency table, and provider logs document both the null-result boundary conditions and the exact artifacts required to test alternative hypotheses (e.g., higher task ambiguity, stricter validators, adversarial stress generators, multi-model comparisons). Because the preregistration and execution artifacts are public within this evidence bundle, researchers can systematically vary one factor at a time

and quantify how discordant counts and the paired-difference distribution change — an approach that will be necessary to move from context-specific null findings toward generalizable operational guidance about when verifier scaffolding materially improves NetOps safety.

VI. LIMITATIONS

- Single model family tested (gpt-5-mini); results do not imply generality across other LLMs or fine-tuned variants.
- Task corpus skew: synthetic/consented tasks yielded near-ceiling baseline success ($\approx 99.75\%$), limiting headroom and statistical power to detect practical improvements by guarded pipelines.
- Emulator fidelity: Mininet-style emulation and deterministic validators approximate production behavior but may not capture all deployment failure modes (routing convergence, vendor idiosyncrasies, transient telemetry).
- Verifier configuration dependence: deterministic_netops_validator_v5 acceptance thresholds and parsing rules materially affect outcomes; different verifier strictness could change repair invocation rates and measured gains.
- Analytic deviation documented: the preregistration specified a Wilcoxon signed-rank paired procedure; the archived confirmatory analysis used an exact McNemar test because outcomes were binary. This deviation is recorded in the archived analysis artifacts and in the Disclosure Statement above.
- Operational external validity: absence of heterogeneous vendor renderers, real production templates, and live rollouts constrain direct production generalization.

VII. CONCLUSION

We preregistered and executed NetOpsTraceBench, a paired, emulator-grounded study comparing one-shot and verifier-assisted NL $\rightarrow$ configuration pipelines on 1,199 completed pairs. Under our synthetic task corpus, deterministic validator settings, and gpt-5-mini as requested, baseline one-shot generation already achieved near-ceiling success ($\approx 99.75\%$). The guarded pipeline increased absolute success by ≈ 0.25 percentage points (paired difference = 0.0025020851) with exact McNemar two-sided $p = 0.25$ and a 95% bootstrap CI that includes zero. Archived execution manifests, provider audits (model_calls_used = 1,203; repairs invoked = 3), representative validator traces, and analysis artifacts support the interpretation that the observed null result reflects limited headroom given task and verifier choices rather than definitive evidence that verifiers are never useful. Future work should intentionally design more challenging and adversarial task sets, vary verifier strictness, and test multiple LLM families to determine conditions under which verifier scaffolding yields operationally meaningful improvements. All run artifacts, analysis code, seeds, and logs are archived to enable exact reproduction and follow-up sensitivity studies (see Disclosure Statement).

DISCLOSURE STATEMENT

Mandatory disclosure (preregistered run rX4f7-1). LLM(s) and provider: requested model = gpt-5-mini via provider recorded as openai_responses; model configuration archived at execution/model_configuration.json (requested_model=gpt-5-mini, max_output_tokens=1500). Orchestration/adaptor: hosted_netops_gvr_v1 executed the paired benchmark. Deterministic tooling: deterministic_netops_validator_v5 performed canonical parsing, intent-constraint validation, and emulator preflight checks; deterministic_netops_task_generator_v5 produced synthetic tasks. Exact initial master prompt: archived at provenance/master_prompt.txt (immutable reference SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acd1582bb39b15ba). Topic constraint and autonomy boundary: the human Research Director selected the topic family and approved the master prompt pre-lock; after the autonomy lock the AI autonomously formulated the research question, conducted literature review, generated the preregistration, executed the experiment, performed analysis, and wrote and revised the manuscript. Preregistration: NetOpsTraceBench (ID rX4f7-1) preregistration manifest SHA-256 = 77abe8b493bde44484e0b3d02cfd7f9ec1621ff7e7847889082ae177a7ed5712; preregistered design (confirmatory hypotheses, inclusion/exclusion, seeds, stopping rules, analysis plan) was frozen before execution. Execution summary: started 2026-08-30T01:21:55.415955+00:00, completed 2026-08-30T03:15:12.141890+00:00; planned episodes 2,400; completed episodes 2,398; completed pairs 1,199; model_calls_used = 1,203; repair-stage calls observed = 3. Analysis artifacts and deviation record: the preregistration specified a Wilcoxon signed-rank paired procedure; the archived executed confirmatory analysis used an exact McNemar two-sided test on paired binary outcomes plus a paired bootstrap percentile CI. This post-preregistration analytic choice is documented in the archived analysis artifacts (analysis/paired_contingency_table.csv, analysis/results.json, analysis/analysis_log.jsonl, analysis/paired_difference_figure.svg) produced as part of the run that completed at 2026-08-30T03:15:12.141890+00:00. Artifacts and reproducibility: raw model outputs, provider traces, validator traces, task_manifest, transformation_manifest, execution logs, analysis code, and all seeds are archived; artifact_count = 8,408 and full hash manifest path = execution/execution_manifest.json. Human involvement: pre-lock collaboration and infrastructure development occurred; no human manual editing, selective revision, or ad hoc text insertion occurred on the manuscript content after the autonomy lock—the AI performed internal autonomous revision cycles only. Technical restarts and deviations: execution manifest records two failed episodes and the analytic deviation described above; all deviations are logged in execution/execution_log.jsonl. The disclosure above is complete relative to the archived artifacts supplied with this run.

REFERENCES

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